

eight tries, most people will assume the problem is with the key and will go back to the front desk to get a new one. One way to understand this decision is to say that the initial problem formulation (observable, nondeterministic) is abandoned in favor of a different formulation (partially observable, deterministic) where the failure is attributed to an unobservable property of the key. We have more to say on this issue in Chapter 13.

4.4 SEARCHING WITH PARTIAL OBSERVATIONS

BELIEF STATE

We now turn to the problem of partial observability, where the agent's percepts do not suffice to pin down the exact state. As noted at the beginning of the previous section, if the agent is in one of several possible states, then an action may lead to one of several possible outcomes—even if the environment is deterministic. The key concept required for solving partially observable problems is the **belief state**, representing the agent's current belief about the possible physical states it might be in, given the sequence of actions and percepts up to that point. We begin with the simplest scenario for studying belief states, which is when the agent has no sensors at all; then we add in partial sensing as well as nondeterministic actions.

4.4.1 Searching with no observation

SENSORLESS

CONFORMANT

When the agent's percepts provide *no information at all*, we have what is called a **sensorless** problem or sometimes a **conformant** problem. At first, one might think the sensorless agent has no hope of solving a problem if it has no idea what state it's in; in fact, sensorless problems are quite often solvable. Moreover, sensorless agents can be surprisingly useful, primarily because they *don't* rely on sensors working properly. In manufacturing systems, for example, many ingenious methods have been developed for orienting parts correctly from an unknown initial position by using a sequence of actions with no sensing at all. The high cost of sensing is another reason to avoid it: for example, doctors often prescribe a broad-spectrum antibiotic rather than using the contingent plan of doing an expensive blood test, then waiting for the results to come back, and then prescribing a more specific antibiotic and perhaps hospitalization because the infection has progressed too far.

COERCION

We can make a sensorless version of the vacuum world. Assume that the agent knows the geography of its world, but doesn't know its location or the distribution of dirt. In that case, its initial state could be any element of the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Now, consider what happens if it tries the action *Right*. This will cause it to be in one of the states $\{2, 4, 6, 8\}$ —the agent now has more information! Furthermore, the action sequence $[Right, Suck]$ will always end up in one of the states $\{4, 8\}$. Finally, the sequence $[Right, Suck, Left, Suck]$ is guaranteed to reach the goal state 7 no matter what the start state. We say that the agent can **coerce** the world into state 7.

To solve sensorless problems, we search in the space of belief states rather than physical states.¹⁰ Notice that in belief-state space, the problem is *fully observable* because the agent

¹⁰ In a fully observable environment, each belief state contains one physical state. Thus, we can view the algorithms in Chapter 3 as searching in a belief-state space of singleton belief states.

always knows its own belief state. Furthermore, the solution (if any) is always a sequence of actions. This is because, as in the ordinary problems of Chapter 3, the percepts received after each action are completely predictable—they're always empty! So there are no contingencies to plan for. This is true *even if the environment is nondeterministic*.

It is instructive to see how the belief-state search problem is constructed. Suppose the underlying physical problem P is defined by ACTIONS_P , RESULT_P , GOAL-TEST_P , and STEP-COST_P . Then we can define the corresponding sensorless problem as follows:

- **Belief states:** The entire belief-state space contains every possible set of physical states. If P has N states, then the sensorless problem has up to 2^N states, although many may be unreachable from the initial state.
- **Initial state:** Typically the set of all states in P , although in some cases the agent will have more knowledge than this.
- **Actions:** This is slightly tricky. Suppose the agent is in belief state $b = \{s_1, s_2\}$, but $\text{ACTIONS}_P(s_1) \neq \text{ACTIONS}_P(s_2)$; then the agent is unsure of which actions are legal. If we assume that illegal actions have no effect on the environment, then it is safe to take the *union* of all the actions in any of the physical states in the current belief state b :

$$\text{ACTIONS}(b) = \bigcup_{s \in b} \text{ACTIONS}_P(s) .$$

On the other hand, if an illegal action might be the end of the world, it is safer to allow only the *intersection*, that is, the set of actions legal in *all* the states. For the vacuum world, every state has the same legal actions, so both methods give the same result.

- **Transition model:** The agent doesn't know which state in the belief state is the right one; so as far as it knows, it might get to any of the states resulting from applying the action to one of the physical states in the belief state. For deterministic actions, the set of states that might be reached is

$$b' = \text{RESULT}(b, a) = \{s' : s' = \text{RESULT}_P(s, a) \text{ and } s \in b\} . \quad (4.4)$$

With deterministic actions, b' is never larger than b . With nondeterminism, we have

$$\begin{aligned} b' = \text{RESULT}(b, a) &= \{s' : s' \in \text{RESULTS}_P(s, a) \text{ and } s \in b\} \\ &= \bigcup_{s \in b} \text{RESULTS}_P(s, a) , \end{aligned}$$

which may be larger than b , as shown in Figure 4.13. The process of generating the new belief state after the action is called the **prediction** step; the notation $b' = \text{PREDICT}_P(b, a)$ will come in handy.

- **Goal test:** The agent wants a plan that is sure to work, which means that a belief state satisfies the goal only if *all* the physical states in it satisfy GOAL-TEST_P . The agent may *accidentally* achieve the goal earlier, but it won't *know* that it has done so.
- **Path cost:** This is also tricky. If the same action can have different costs in different states, then the cost of taking an action in a given belief state could be one of several values. (This gives rise to a new class of problems, which we explore in Exercise 4.9.) For now we assume that the cost of an action is the same in all states and so can be transferred directly from the underlying physical problem.

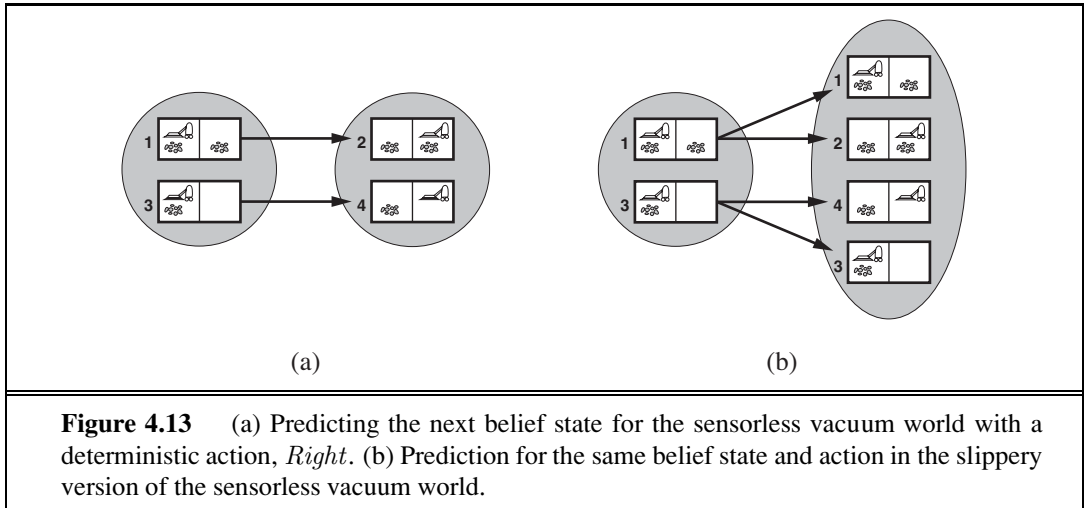


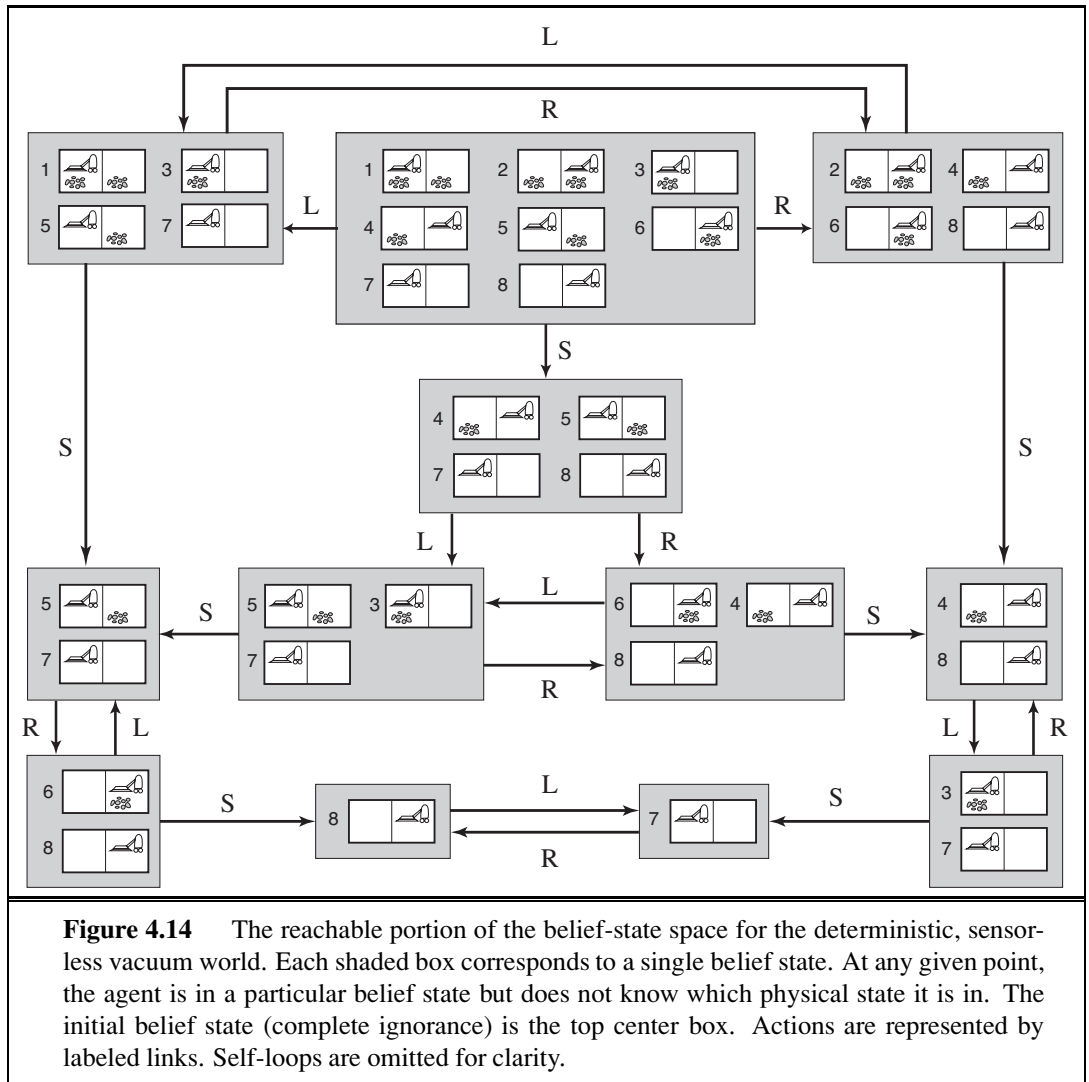
Figure 4.13 (a) Predicting the next belief state for the sensorless vacuum world with a deterministic action, *Right*. (b) Prediction for the same belief state and action in the slippery version of the sensorless vacuum world.

Figure 4.14 shows the reachable belief-state space for the deterministic, sensorless vacuum world. There are only 12 reachable belief states out of $2^8 = 256$ possible belief states.

The preceding definitions enable the automatic construction of the belief-state problem formulation from the definition of the underlying physical problem. Once this is done, we can apply any of the search algorithms of Chapter 3. In fact, we can do a little bit more than that. In “ordinary” graph search, newly generated states are tested to see if they are identical to existing states. This works for belief states, too; for example, in Figure 4.14, the action sequence [*Suck*,*Left*,*Suck*] starting at the initial state reaches the same belief state as [*Right*,*Left*,*Suck*], namely, $\{5, 7\}$. Now, consider the belief state reached by [*Left*], namely, $\{1, 3, 5, 7\}$. Obviously, this is not identical to $\{5, 7\}$, but it is a *superset*. It is easy to prove (Exercise 4.8) that if an action sequence is a solution for a belief state b , it is also a solution for any subset of b . Hence, we can discard a path reaching $\{1, 3, 5, 7\}$ if $\{5, 7\}$ has already been generated. Conversely, if $\{1, 3, 5, 7\}$ has already been generated and found to be solvable, then any *subset*, such as $\{5, 7\}$, is guaranteed to be solvable. This extra level of pruning may dramatically improve the efficiency of sensorless problem solving.

Even with this improvement, however, sensorless problem-solving as we have described it is seldom feasible in practice. The difficulty is not so much the vastness of the belief-state space—even though it is exponentially larger than the underlying physical state space; in most cases the branching factor and solution length in the belief-state space and physical state space are not so different. The real difficulty lies with the size of each belief state. For example, the initial belief state for the 10×10 vacuum world contains 100×2^{100} or around 10^{32} physical states—far too many if we use the atomic representation, which is an explicit list of states.

One solution is to represent the belief state by some more compact description. In English, we could say the agent knows “Nothing” in the initial state; after moving *Left*, we could say, “Not in the rightmost column,” and so on. Chapter 7 explains how to do this in a formal representation scheme. Another approach is to avoid the standard search algorithms, which treat belief states as black boxes just like any other problem state. Instead, we can look



inside the belief states and develop **incremental belief-state search** algorithms that build up the solution one physical state at a time. For example, in the sensorless vacuum world, the initial belief state is $\{1, 2, 3, 4, 5, 6, 7, 8\}$, and we have to find an action sequence that works in all 8 states. We can do this by first finding a solution that works for state 1; then we check if it works for state 2; if not, go back and find a different solution for state 1, and so on. Just as an AND–OR search has to find a solution for every branch at an AND node, this algorithm has to find a solution for every state in the belief state; the difference is that AND–OR search can find a different solution for each branch, whereas an incremental belief-state search has to find *one* solution that works for *all* the states.

The main advantage of the incremental approach is that it is typically able to detect failure quickly—when a belief state is unsolvable, it is usually the case that a small subset of the belief state, consisting of the first few states examined, is also unsolvable. In some cases,

this leads to a speedup proportional to the size of the belief states, which may themselves be as large as the physical state space itself.

Even the most efficient solution algorithm is not of much use when no solutions exist. Many things just cannot be done without sensing. For example, the sensorless 8-puzzle is impossible. On the other hand, a little bit of sensing can go a long way. For example, every 8-puzzle instance is solvable if just one square is visible—the solution involves moving each tile in turn into the visible square and then keeping track of its location.

4.4.2 Searching with observations

For a general partially observable problem, we have to specify how the environment generates percepts for the agent. For example, we might define the local-sensing vacuum world to be one in which the agent has a position sensor and a local dirt sensor but has no sensor capable of detecting dirt in other squares. The formal problem specification includes a $\text{PERCEPT}(s)$ function that returns the percept received in a given state. (If sensing is nondeterministic, then we use a PERCEPTS function that returns a set of possible percepts.) For example, in the local-sensing vacuum world, the PERCEPT in state 1 is $[A, \text{Dirty}]$. Fully observable problems are a special case in which $\text{PERCEPT}(s) = s$ for every state s , while sensorless problems are a special case in which $\text{PERCEPT}(s) = \text{null}$.

When observations are partial, it will usually be the case that several states could have produced any given percept. For example, the percept $[A, \text{Dirty}]$ is produced by state 3 as well as by state 1. Hence, given this as the initial percept, the initial belief state for the local-sensing vacuum world will be $\{1, 3\}$. The ACTIONS , STEP-COST , and GOAL-TEST are constructed from the underlying physical problem just as for sensorless problems, but the transition model is a bit more complicated. We can think of transitions from one belief state to the next for a particular action as occurring in three stages, as shown in Figure 4.15:

- The **prediction** stage is the same as for sensorless problems: given the action a in belief state b , the predicted belief state is $\hat{b} = \text{PREDICT}(b, a)$.¹¹
- The **observation prediction** stage determines the set of percepts o that could be observed in the predicted belief state:

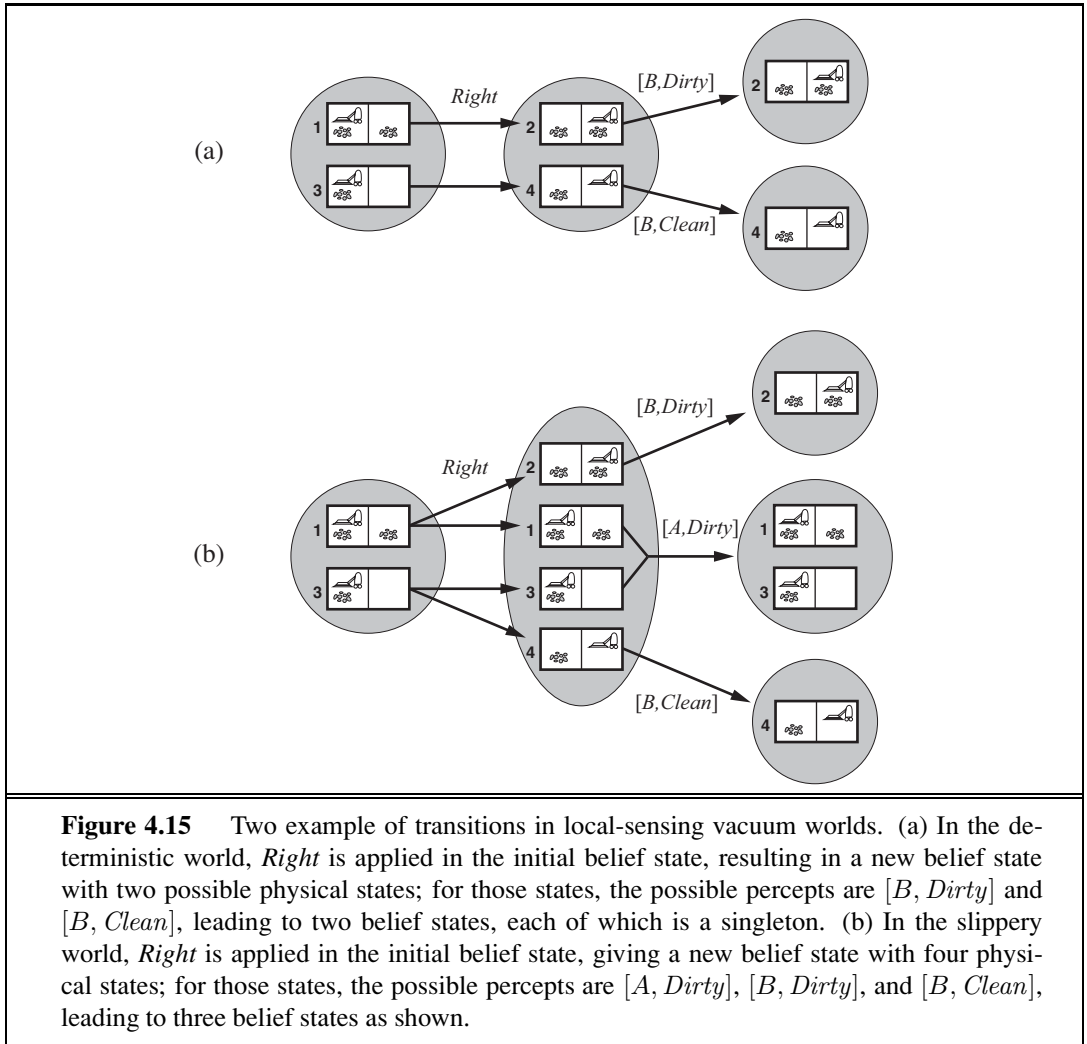
$$\text{POSSIBLE-PERCEPTS}(\hat{b}) = \{o : o = \text{PERCEPT}(s) \text{ and } s \in \hat{b}\}.$$

- The **update** stage determines, for each possible percept, the belief state that would result from the percept. The new belief state b_o is just the set of states in $\hat{b}$ that could have produced the percept:

$$b_o = \text{UPDATE}(\hat{b}, o) = \{s : o = \text{PERCEPT}(s) \text{ and } s \in \hat{b}\}.$$

Notice that each updated belief state b_o can be no larger than the predicted belief state $\hat{b}$; observations can only help reduce uncertainty compared to the sensorless case. Moreover, for deterministic sensing, the belief states for the different possible percepts will be disjoint, forming a *partition* of the original predicted belief state.

¹¹ Here, and throughout the book, the “hat” in $\hat{b}$ means an estimated or predicted value for b .



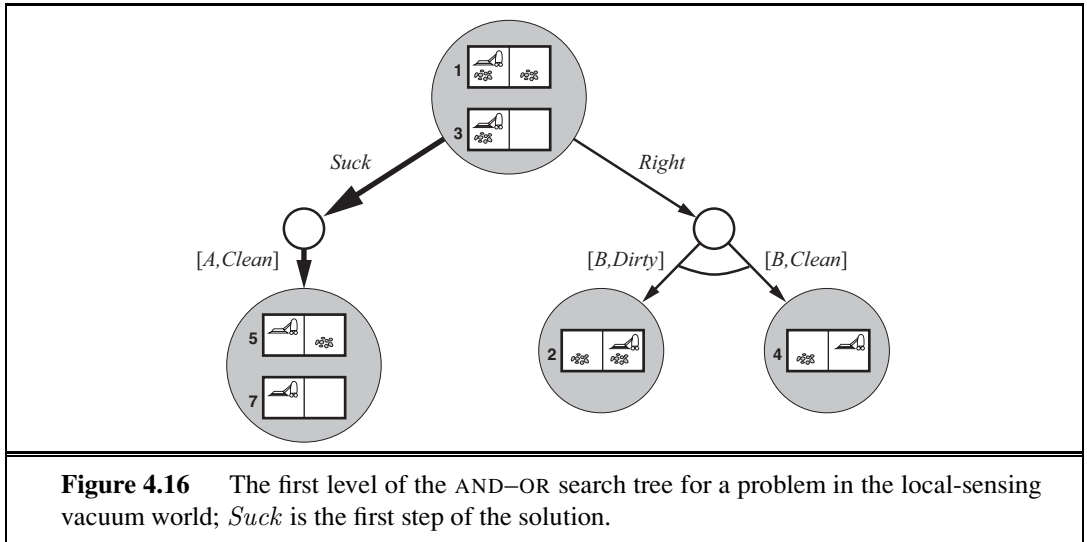
Putting these three stages together, we obtain the possible belief states resulting from a given action and the subsequent possible percepts:

$$\begin{aligned} \text{RESULTS}(b, a) = \{b_o : b_o = & \text{UPDATE}(\text{PREDICT}(b, a), o) \text{ and} \\ & o \in \text{POSSIBLE-PERCEPTS}(\text{PREDICT}(b, a))\}. \end{aligned} \quad (4.5)$$

Again, the nondeterminism in the partially observable problem comes from the inability to predict exactly which percept will be received after acting; underlying nondeterminism in the physical environment may *contribute* to this inability by enlarging the belief state at the prediction stage, leading to more percepts at the observation stage.

4.4.3 Solving partially observable problems

The preceding section showed how to derive the RESULTS function for a nondeterministic belief-state problem from an underlying physical problem and the PERCEPT function. Given



such a formulation, the AND-OR search algorithm of Figure 4.11 can be applied directly to derive a solution. Figure 4.16 shows part of the search tree for the local-sensing vacuum world, assuming an initial percept $[A, \text{Dirty}]$. The solution is the conditional plan

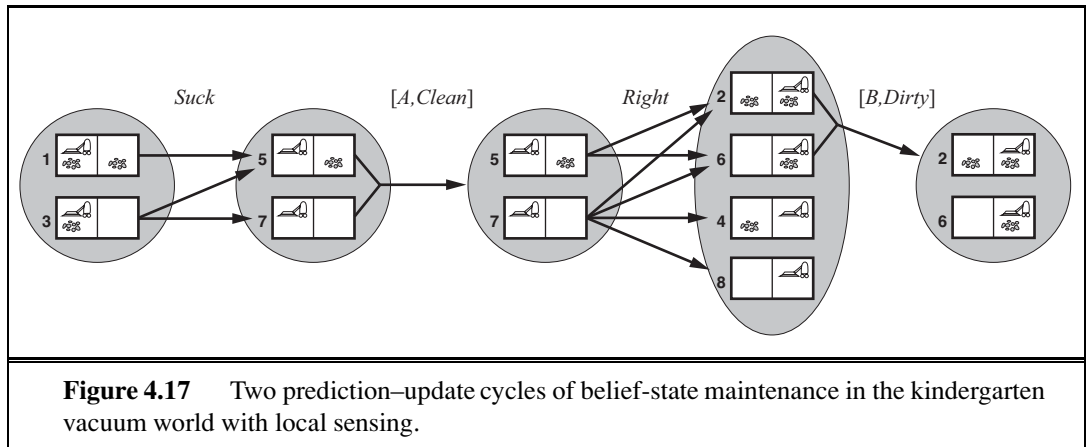
$[Suck, Right, \text{if } Bstate = \{6\} \text{ then } Suck \text{ else } []]$.

Notice that, because we supplied a belief-state problem to the AND-OR search algorithm, it returned a conditional plan that tests the belief state rather than the actual state. This is as it should be: in a partially observable environment the agent won't be able to execute a solution that requires testing the actual state.

As in the case of standard search algorithms applied to sensorless problems, the AND-OR search algorithm treats belief states as black boxes, just like any other states. One can improve on this by checking for previously generated belief states that are subsets or supersets of the current state, just as for sensorless problems. One can also derive incremental search algorithms, analogous to those described for sensorless problems, that provide substantial speedups over the black-box approach.

4.4.4 An agent for partially observable environments

The design of a problem-solving agent for partially observable environments is quite similar to the simple problem-solving agent in Figure 3.1: the agent formulates a problem, calls a search algorithm (such as AND-OR-GRAPH-SEARCH) to solve it, and executes the solution. There are two main differences. First, the solution to a problem will be a conditional plan rather than a sequence; if the first step is an if-then-else expression, the agent will need to test the condition in the if-part and execute the then-part or the else-part accordingly. Second, the agent will need to maintain its belief state as it performs actions and receives percepts. This process resembles the prediction-observation-update process in Equation (4.5) but is actually simpler because the percept is given by the environment rather than calculated by the



agent. Given an initial belief state b , an action a , and a percept o , the new belief state is:

$$b' = \text{UPDATE}(\text{PREDICT}(b, a), o) . \quad (4.6)$$

Figure 4.17 shows the belief state being maintained in the *kindergarten* vacuum world with local sensing, wherein any square may become dirty at any time unless the agent is actively cleaning it at that moment.¹²

In partially observable environments—which include the vast majority of real-world environments—maintaining one’s belief state is a core function of any intelligent system. This function goes under various names, including **monitoring**, **filtering** and **state estimation**. Equation (4.6) is called a **recursive** state estimator because it computes the new belief state from the previous one rather than by examining the entire percept sequence. If the agent is not to “fall behind,” the computation has to happen as fast as percepts are coming in. As the environment becomes more complex, the exact update computation becomes infeasible and the agent will have to compute an approximate belief state, perhaps focusing on the implications of the percept for the aspects of the environment that are of current interest. Most work on this problem has been done for stochastic, continuous-state environments with the tools of probability theory, as explained in Chapter 15. Here we will show an example in a discrete environment with deterministic sensors and nondeterministic actions.

The example concerns a robot with the task of **localization**: working out where it is, given a map of the world and a sequence of percepts and actions. Our robot is placed in the maze-like environment of Figure 4.18. The robot is equipped with four sonar sensors that tell whether there is an obstacle—the outer wall or a black square in the figure—in each of the four compass directions. We assume that the sensors give perfectly correct data, and that the robot has a correct map of the environment. But unfortunately the robot’s navigational system is broken, so when it executes a *Move* action, it moves randomly to one of the adjacent squares. The robot’s task is to determine its current location.

Suppose the robot has just been switched on, so it does not know where it is. Thus its initial belief state b consists of the set of all locations. The the robot receives the percept

¹² The usual apologies to those who are unfamiliar with the effect of small children on the environment.

MONITORING

FILTERING

STATE ESTIMATION

RECURSIVE

LOCALIZATION

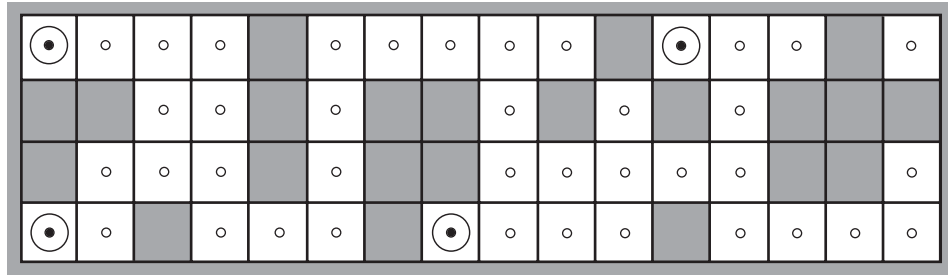
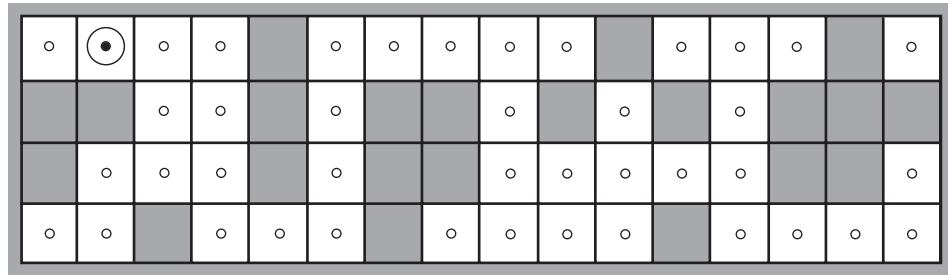
(a) Possible locations of robot after $E_1 = NSW$ (b) Possible locations of robot After $E_1 = NSW, E_2 = NS$

Figure 4.18 Possible positions of the robot, $\odot$, (a) after one observation $E_1 = NSW$ and (b) after a second observation $E_2 = NS$. When sensors are noiseless and the transition model is accurate, there are no other possible locations for the robot consistent with this sequence of two observations.

NSW, meaning there are obstacles to the north, west, and south, and does an update using the equation $b_o = \text{UPDATE}(b)$, yielding the 4 locations shown in Figure 4.18(a). You can inspect the maze to see that those are the only four locations that yield the percept *NSW*.

Next the robot executes a *Move* action, but the result is nondeterministic. The new belief state, $b_a = \text{PREDICT}(b_o, \text{Move})$, contains all the locations that are one step away from the locations in b_o . When the second percept, *NS*, arrives, the robot does $\text{UPDATE}(b_a, NS)$ and finds that the belief state has collapsed down to the single location shown in Figure 4.18(b). That's the only location that could be the result of

$$\text{UPDATE}(\text{PREDICT}(\text{UPDATE}(b, NSW), \text{Move}), NS) .$$

With nondeterministic actions the *PREDICT* step grows the belief state, but the *UPDATE* step shrinks it back down—as long as the percepts provide some useful identifying information. Sometimes the percepts don't help much for localization: If there were one or more long east-west corridors, then a robot could receive a long sequence of *NS* percepts, but never know where in the corridor(s) it was.